

Guided Lab — Course Registration Database

Week 8 • Day 3 — Database and RDBMS Foundations

Design
Plan
Understand
Build



1 Identify the Student, Course, and Enrollment entities

Three entities are needed to model course registration:

- **Student** — a person who can register for courses.
- **Course** — a class that students can register for.
- **Enrollment** — the record of one student registering for one course.

Enrollment is its own entity, not just a field inside Student or Course, because a student can register for many courses and a course can have many students. Enrollment describes the link between them, not a property of either one alone.



2 Give each entity a primary key

Every table needs a **Primary Key**: a column that uniquely identifies each row, cannot repeat, and cannot be empty. The naming pattern used here is `<entity>_id`.

- Student → **student_id**
- Course → **course_id**
- Enrollment → **enrollment_id**

Example: two students can both be named "Sara," but each gets a different `student_id`, so the database can always tell them apart.



3 Add three useful attributes to Student and Course

Attributes beyond the primary key describe each entity:

- Student: name, email, program
- Course: title, capacity, instructor



Attributes describe each entity.

4 Decide which foreign keys belong in Enrollment

Enrollment links Student and Course without repeating their data. It stores a Foreign Key for each — the primary key of the related record, not the record's full data.

- **enrollment_id** (Primary Key)
- **student_id** (Foreign Key → Student.**student_id**)
- **course_id** (Foreign Key → Course.**course_id**)
- **enrolled_at** (date the registration happened)



Foreign Keys connect tables.

5 State the relationship between Student and Enrollment

One student can have many enrollments.

A student can register for more than one course, so each student can be linked to multiple enrollment records. This is a **one-to-many** relationship.



Student (1)
→ Enrollment (Many)

6 State the relationship between Course and Enrollment

One course can have many enrollments.

A course can have many students registered in it, so each course can also be linked to multiple enrollment records. Together, Steps 5 and 6 show why **Enrollment** exists: it breaks the **many-to-many** relationship between Student and Course into two simple **one-to-many** relationships.



Course (1)
→ Enrollment (Many)

7 Add one UNIQUE rule and one NOT NULL rule

- **UNIQUE** → Student.**email** — no two students can share the same email address.
- **NOT NULL** → Course.**title** — every course record must have a title; it cannot be left empty.

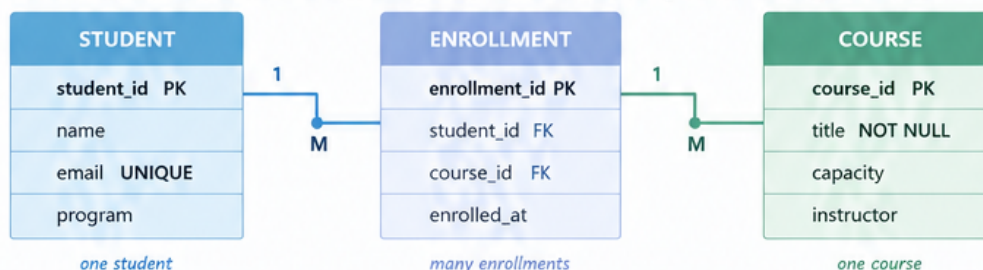
Note: `student_id` and `course_id` are already **unique** by definition because they are Primary Keys, so a separate UNIQUE rule is *not* needed there. The rules above apply to ordinary attributes instead.



Data Rules
keep the data clean
and consistent.

8 Draw the three tables and connect their keys

The diagram below shows all three tables with explicit **1-to-M** connector lines, matching standard ER diagram notation.



Note: Attribute names (*program*, *enrolled_at*, *capacity*, *instructor*) and the specific NOT NULL field are design choices. Other reasonable choices are valid as long as the same relational logic — **one primary key per table**, **foreign keys pointing to real records**, and **no unnecessary repetition** — is followed.

Exit Ticket

Why should Enrollment not store the student's name and course title repeatedly?

Enrollment does not store the student's name or the course title directly. Instead it stores Foreign Keys (`student_id`, `course_id`) that point to the original records in Students and Courses. If the name and title were copied into every enrollment row, changing a course title later would require updating every matching enrollment record by hand — slow with many rows, and risky, because a missed update leaves the data inconsistent (two rows disagreeing about the same course). Using a Foreign Key means the value is changed once, in the original table, and every enrollment that references it automatically reflects the update. This keeps a single source of truth for each fact.

Keep
Going